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A data-driven approach to customer lifetime value prediction using probability and machine learning models

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ABSTRACT

Customer lifetime value is an important marketing metric and has applications in market segmentation, strategy development, and direct marketing programs, especially when customers are not under contract. In this research, we demonstrate the prediction of the lifetime value of patients in a health service portfolio in two separate ways. The probability of a patient being alive and their value in the coming evaluation period are first predicted using a probability model that has been well-established in the marketing community. We then use several machine learning algorithms to perform the same task. The results of these two approaches are compared in terms of accuracy to gain insight into their respective strengths and weaknesses. We believe that the work is one of the first attempts to gain an understanding of the use of machine learning algorithms in this important marketing issue. The results showed that the probability model performs better than the machine learning models, probably due to the assumption required in the probability calculations. It is therefore recommended that an essential step in applying these software approaches is to verify the validity of the key assumption of regularity. In addition, in future studies, consideration should be given to a larger dataset with demographic variables beyond age and gender that were used in this study. Developing specific ML models for dealing with zero-inflated data, which is an inherent feature of customer lifetime data, will also be helpful.

1. Introduction

As we move through the information revolution, new marketing challenges are coming up that require more research. Issues, new and old, can be looked at in better ways because we have access to better data and predictive analytics. An area of marketing that benefits from this change is the customer lifetime value (CLV).

The growing importance of a customer-focused approach in marketing, such as one-to-one marketing and database marketing, is clear from the many customer relationship management initiatives we see today. These initiatives aim to extend the customer life cycle and increase the value of transactions between the company and customers. Businesses are now prioritizing the development of long-lasting customer relationships to achieve higher service quality and growth. As a result, marketing activities and performance assessments are increasingly based on customer relationships rather than products [1]. This shift has created a new way to make and evaluate marketing decisions. Understanding CLV is crucial in this context, as CLV models provide a systematic way to analyze and assess a company's relationship with its customers.

According to Gupta [2], "companies derive revenue from creating and sustaining long-term relationships with their customers. In such an environment, marketing maximizes CLV and customer equity". The CLV of a customer is the predicted net present value of the revenue stream (or profit) that the customer will generate through future transactions. Companies use a metric to "assess a business's customers worth and an accurate metric in measuring the company's potential customers" [3]. CLV is a way to measure the total value that a customer is expected to bring to a business from their relationship. It is an important metric because it helps companies understand which customers are the most valuable and where to focus their efforts.

In general, CLV supports a relationship-based, customer-focused marketing approach [4]. In addition to determining long-term customer values, understanding the CLV structure of a firm helps to balance its customer acquisition and retention costs. CLV also provides insights into the loyalty and engagement of customers and allows for proper market segmentation and in making strategic marketing decisions. For example, through a combined quantitative and qualitative investigation, Ali and Shabn [5] examined the interaction between CLV, strategic marketing practices, and their impact on organizational financial performance. The results of the study showed the pivotal role of CLV in influencing financial metrics such as market valuation (P/E ratio), as well as positive customer perceptions regarding CLV factors.

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From a research perspective, Abdolvand et al. [6] performed a critical analysis of the CLV literature and offered a comprehensive model to increase the applicability of CLV, as well as the need for more research on the implementation and validation of CLV. A detailed literature review on the importance of CLV in modern marketing can be found in [7].

As a metric, CLV is helpful for market segmentation [8], for the evaluation of the effectiveness of marketing programs [9], as well as for the allocation of marketing resources for customer acquisition, retention, and expansion (cross-selling and up-selling). The CLV could also help evaluate whether a company's customer relationship management system (CRM) effectively achieves its strategic objectives [10,11]. The primary use is in the segmentation of a customer base. For example, Dahana et al. [12] linked CLV and lifestyle for an online fashion retail market and produced four customer segments that were different in average CLV. Similarly, Kanchanapoom and Chongwatpol [8] integrated CLV and customer migration and created modified segmentation models that were validated "in the context of complementary and alternative medicine in the healthcare industry to demonstrate the practical validity of our proposed methods". Other examples of the use of CLV in marketing segmentation and targeting can be found in von Mutius and Huchzermeier [13], and Binh et al. [14], and Yang [15].

Fader et al. [16] state that businesses often undervalue customers by relying on short-term metrics, such as Customer Acquisition Cost (CPA), which prioritizes cost cutting over strategic investment in high-value customers. Instead, Fader et al. [16] argue that CLV should be the guiding metric because it evaluates the long-term profitability of customer relationships. They explain that customer equity, the aggregate CLV of current and future customers, represents a firm's most significant asset. For an organization, this means prioritizing customers that are likely to provide sustained engagement and growth.

Customers in a service business could be under contract. The current and future revenue streams produced by these customers could be projected with a high level of certainty and regularity. However, many customers in a transactional relationship are not under contract. The calculations and modeling of their lifetime values are very different as they could terminate the relationship with the business at any time, usually without the knowledge of the business. Many research projects have been devoted to estimating the CLV in this scenario. See, for example, Fader et al. [16], Marmol et al. [17], Mudjahidin et al. [18], Dogan et al. [19].

In this paper, we will present two different sets of algorithms to predict CLV, using a sample data set from a health service portfolio. First, the probability that a customer (patient) is alive and their value in the evaluation period is calculated using a probability model established by Fader et al. [16]. We will then use several machine learning (ML) algorithms to try and accomplish the same task. The results of these two approaches will then be compared in terms of accuracy. We believe that the work is one of the first attempts to gain an understanding of the use of ML in this important marketing issue.

The main contribution of this research is to demonstrate the use of probability and machine learning algorithms as ways to predict the customer life-time values and determine whether a patient is "alive" under the Buy Till You Die (BTYD) paradigm. More importantly, results from the two computational and modeling approaches are compared to gain insight into their strengths and weaknesses and to inform the potential use of these algorithms in practice.

In the following presentation, Section 2 presents a comprehensive review of the literature on the development of CLV calculation and modeling. Section 3 presents the data used for this research. Sections 4 and 5 present the results of applying the probability and machine learning algorithms to the sample data set. Sections 6 and 7 compare these results and give recommendations on the possible use of these algorithms. This paper closes with a brief conclusion in Section 8.

2. Literature review

A logical way to predict the lifetime value of a customer is to study the customer's purchase patterns, which is called a customer base analysis in the literature. Ehrenberg [20] showed how negative binomial distribution (NBD) can be used to model data on purchases of nondurable consumer goods. The author also considered a simple statistical model for purchases made over time. This model served as a reference point for those developed later in a heterogeneous purchase process. Note that the NBD model did not account for the impact of customer churn.

To estimate future purchasing patterns of customers and use them to predict their lifetime value, Schmittlein et al. [21] proposed the Pareto/NBD "counting your customers" framework in 1987. The model added a heterogeneous dropout process but was found to be difficult to implement due to the computational challenges associated with parameter estimation. Fader et al. [22] developed a new BG/NBD (beta-geometric/NBD) model that was "a slight variation in the behavioral story associated with Pareto/NBD but is vastly easier to implement". The authors showed that the parameters could be easily estimated using Microsoft Excel. More importantly, their model yields very similar results in various purchasing environments. The purchasing condition is the same in both models. Customers can churn randomly without contractual restrictions. Both models also allow for customer heterogeneity: customers may differ in purchase and dropout behaviors. One advantage of these models is that the only purchase data required are recency and frequency.

An extension of the ideas in Schmittlein et al. [21] was proposed in 2007 by Ma and Liu [23]. In this paper, the authors proposed a Markov Chain Monte Carlo (MCMC) algorithm to estimate the parameters of the Pareto/NBD model. The outcome of this approach was that the estimates are expressed as probability distributions and, therefore, interval estimates could also be constructed.

Other extensions of the above work included the Modified BG/NBD (MBG/NBD) model where a customer could drop out at time zero with a positive probability [24,25]. In other words, this model removed inconsistencies with the BG/NBD model by allowing customers without any activity to remain inactive. Abe in 2020 [26] proposed a customer CLV model that accommodated memoryless or cyclical purchase patterns. This was to overcome the restrictive assumption made in the Pareto/NBD model of Poisson purchase, which imposed a memoryless purchase process and the purchase occurred independently of the previous inter-purchase time.

Fader et al. [16] applied the Pareto/NBD concept and developed a model of future purchasing patterns for a customer base in a non-contractual discrete-time setting. This BG/BB (beta-geometric/beta-Bernoulli) model captured "both of the underlying behavioral processes (i.e., customers' purchasing while active and time until each customer dies.)" Again, the model was computationally friendly and gave simple closed-form expressions that can be implemented in Excel for the expected number of future transactions.

The Pareto/NBD model was further extended in 2016 by Platzer and Reutterer [27] through a Pareto/GammaGammaGamma (Pareto/GGG) model by integrating regularity in interpurchase times. By definition, regularity implied less variation in purchase patterns and should allow for better prediction. This was especially important in noncontractual settings as it improved inferences about the status of latent activity of customers. The generalized Pareto/NBD model proposed by the authors accounted for varying degrees of regularity between customers by replacing the NBD component with a mixture of gamma distributions. As expected, this research demonstrated that it was possible to improve customer-level predictions even for mildly regular purchase patterns. The stronger the regularity, the better the prediction. In addition to the recency and frequency statistics, an additional summary statistic on regularity was added to build the model. As the authors pointed out, the cost of this requirement should be minimal.

In addition to modeling the regularity of customer transactions, Reutterer et al. [28] expanded the BG/NBD model by relaxing the assumption of Poisson purchases with a more flexible Erlang k distribution for interpurchase times. According to the authors, the resulting model variants "can leverage regularity while retaining almost the same level of data requirements and algorithmic efficiency". The authors demonstrated improvements in predictive accuracy against baseline models through six datasets and simulation studies. Performance gains were close to or on par with the more complex model alternatives. The implementation of these model variants (BG/CNBD-k and MBG/CNBD-k) was made accessible through the R package BTYDplus [29].

There are other ways to estimate the CLV. Using a hierarchical Bayes approach, Borle et al. [30] estimated the CLV of "each customer at each purchase occasion by jointly modeling the purchase timing, purchase amount, and risk of defection from the firm for each customer". Performance of their model was compared favorably with the extended NBD/Pareto model, the recency, frequency, and monetary (RFM) value model, and a heuristic model that used the average customer lifetime, the average inter-purchase time, and the average dollar purchase amount as input. Their research also showed that longer inter-purchase times were associated with larger purchase amounts and a higher risk of churning. The conclusion on the differences in gender (male versus female) will be incorporated in the machine learning modeling process in this research.

With the proliferation of machine learning (ML) algorithms for classification and prediction tasks, solutions for many business problems in general [31,32] and marketing problems in particular [33–35] have been developed in the last several years. Research projects have also been carried out to segment a customer base [36], determine whether a customer is lost to the business [37], and predict a customer's CLV [38], etc. For example, Akhavan and Hassannayebia [39] proposed a hybrid machine learning framework to predict the outcomes of customers' experiences. Sun et al. [40] conducted a customer segmentation study using ML algorithms and built a segmentation identification model in a non-contractual environment. Bhattacharjee et al. [41] used several machine learning techniques to model the purchasing intention of young Indian consumers for branded smartphones. Delgado [42] used several ML algorithms to predict the CLV of a meal subscription service. Myburg [11] used XGBoost to predict the lifetime value of the customer within the recency, frequency, and monetary (RFM) framework. Other recent work in this area can be found in Asadi Ejgerdi and Kazerooni [43] and Zhao et al. [44].

3. Research data

The data for this study were obtained from a for-profit psychological services firm in Canada. This firm is dedicated to providing mental health assessments and coaching services to uncover hidden strengths and lead more fulfilling lives.

To ensure the security and privacy of patient data, this study focused on extracting transactional information rather than sensitive patient details. Depersonalized transaction data were made available to the researcher team from 5 September 2017 to 20 December 2023, including appointment date, appointment type, duration in minutes, date of birth, and gender. The extracted file has 50,663 records in 4,301 patients.

Under the Recency, Frequency, and Monetary (RFM) paradigm, frequency represents the number of repeat appointments a patient has completed. This means that it is one less than the total number of appointments (page 11 of [45].) Therefore, at least two appointments are required in the complete data set per patient. Only 3,383 patients met these criteria, and their appointments were used to estimate the probability of being alive and to predict the lifetime value.

Typically, a patient comes to the clinic and receives an intake appointment and an assessment as a starting point. Then more coaching appointments will be made, likely regularly, to facilitate the treatment of the identified problems. A small percentage of patients (0.26%) will go directly to treatment. Note that appointment records before September 5, 2017, are not available in the clinic's transaction database. Therefore, data on patients who came to the clinic before that day has been truncated.

Table 1
Number of appointments, Tenure, and Recency.

Statistic	No. appointments	Tenure	Recency
Mean	14.5	1066.1	411.64
St. Dev.	19.26	741.7	496.2
Minimum	2	3	1
First Quartile	5	384.5	70
Median	8	905	210
Third Quartile	16	1837	5595
Maximum	217	2180	2179

The firm's transaction database does not record the fees associated with transactions. Therefore, this critical variable is not available for this research. Instead, we will assume that patients are charged the same consultation fee of $ 200 per hour and that the length of the appointment is sufficient to represent the value of transactions (completed appointment). To conform to the RFM paradigm, we will convert the number of hours into revenue by multiplying the number of hours in an appointment by 200. Note also that age and gender are not used in developing the probability models.

Given the private and for-profit nature of the clinic, we believe that its capacity to provide counseling and consulting services during the study period was not a factor in patient behavior in making and completing appointments. Therefore, this research data is relevant in the estimation of patient lifetime values as long as the clinic manages its operation in a way that is consistent with the past.

Before delving into the study, it is important to note that while the CLV estimation is useful for the private for-profit clinic from a marketing perspective, it may not be the most relevant metric for a health service in general. The treatment and the well-being of its patients should always remain its primary concern.

3.1. Exploratory data analysis

A descriptive analysis of the variables relevant to BD/NBD and related modeling, namely tenure, recency, and regularity, was conducted to gain insight into the data set. Due to the lack of data before September 5, 2017, the first appointment for some patients is not an assessment appointment. Note also that since the frequency statistics generated from the data are underestimates for patients engaged with the firm before 5 September 2017, the predictions produced through this research will also be underestimates. Addressing this issue will require considerable research and will be dealt with in future studies.

Within the RFM framework, tenure is defined as the difference, measured in days, between the first and the last purchase (appointment). Thus, if a patient has one appointment within the study period, the tenure value will be 0. On the other hand, variable recency is defined as the difference between the last transaction date and August 31, 2023, the cutoff date of the transaction file.

With the 3383 patients on file, we summarize the data set for tenure, recency, and frequency (number of repeat appointments) in Table 1.

The variables for calculating the lifetime value were selected after the exploratory data analysis. The patient's depersonalized ID, frequency, recency, and tenure are required for the probability model. Age and sex will be used to develop the machine learning models.

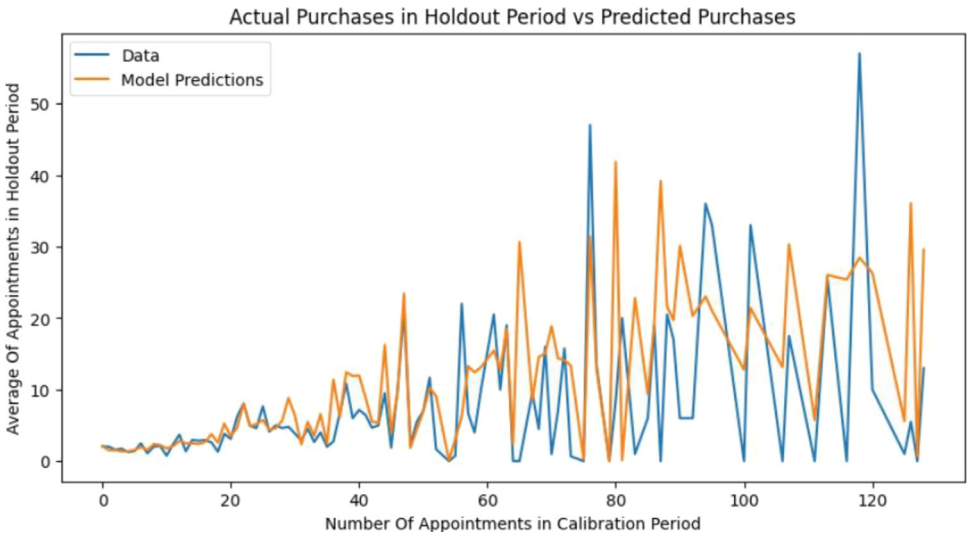

Fig. 1. Number of Appointments in the Calibration and Holdout Period - Actual versus Predicted.

4. Probability models

As mentioned, understanding, evaluating, predicting, and acting on the lifetime value of the customer (CLV) is essential for customer-based marketing efforts [22]. In a series of papers, [46,47] developed several mathematical models (BG/NBD) that were implemented in R [48] and in Excel [16] to evaluate the lifetime value of a customer portfolio. Cameron Davidson-Pilon further implemented these models in 2022 in a Python library called Lifetimes [45,49] and therefore, the prediction of the CLV for a customer portfolio using the probability modeling approach and the BG/NBD paradigm is readily available. This package is comparable to BTYDplus for the purpose of this project, and the choice depends on whether the R or Python environment will be used.

Given that the machine learning algorithms to be implemented for this project are in Python, we will use the data set described above and the Lifetimes package to produce the predictions and compare them with those obtained using the machine learning algorithms.

4.1. The BG/NBD model

In this research, the "BG/NBD" model is used to produce predictions on future transactions, and the Gamma–Gamma model is used to predict the lifetime value of the patients.

The BG/NBD model, created by [22], is a variation of the Pareto/NBD model that was first introduced in [21]. Most of the assumptions that apply to the Pareto/NBD model also apply to the BG/NBD model. An exception is on how and when customers become inactive. The Pareto model assumes that customers can drop out at any time, regardless of whether they have made a purchase. In contrast, the BG model assumes that customers become inactive only after making an actual purchase. We believe that the latter assumption is relevant and valid for the current dataset. In addition, the Pareto/NBD model has challenges in estimating its parameters, leading to computational difficulties. However, the BG model is easier to implement than the Pareto/NBD model.

We made the following assumptions by implementing the BG/NBD model for the research data set described above.

1. As mentioned, given that the actual amount is missing from the data, we calculate the monetary value of transactions (appointments) by multiplying the duration of the appointment (in hours) by $200, the predominant amount charged by the clinic to patients for most types of appointments. This assumption is reasonable for the private clinic.
2. The initiation of recording transactional data for each patient began on 5 September 2017, the start of data extraction.

4.2. Calibration and holdout period

When conducting a customer lifetime value analysis, it is crucial to assess the accuracy of the prediction model compared to the actual behavior of the customer. For this research, the time range in the data set is split into calibration (5 September 2017 to 3 August 2022) and holdout period (4 August 2022 to 31 August 2023). Transactions in the calibration period are used to develop the probability model. Transactions in the holdout period are used to evaluate the accuracy of the model. Note that 82% of the appointment data are in the calibration period and the rest are in the holdout period.

The choice of the cutoff for the calibration period and the holdout period is somewhat arbitrary. In this instance, we choose the date of 4 August 2022 so that approximately 20% of the transaction data is in the holdout period for the probability model, and therefore the same percentage will be in the testing data set for the ML models. Once these two periods are clearly defined, we will be able to evaluate the accuracy of the probability model using data in the holdout period. Likewise, the accuracy of the ML models can also be evaluated using the corresponding testing dataset.

4.3. Predicted number of purchases

This section shows the results of applying the selected probability model to the calibration and holdout period. These outputs are typical when using the BG/NDB model to predict the CLV.

In Fig. 1, the predicted number of appointments during the holdout period is related to the actual number of appointments in the calibration period. In this case, this figure confirms that the model captures the general trend that patients with a high frequency of appointments in the calibration period also have a high frequency in the holdout period. It also shows that the prediction error (actual minus predicted) increases as the number of appointments increases in the calibration period (Fig. 2). This phenomenon makes intuitive sense and is normal in many regression-type modeling exercises. The predictions of the model on the graph are calculated using a simulation to determine the expected number of future purchases [22]. This calculation integrates individual-level expectations over updated distributions of the transaction rate and dropout probability, which are informed by the customer's past purchasing history.

When Fig. 2 is further analyzed, the predicted error is higher when the number of appoint- ments in the calibration period is greater than 33. It should be noted that 8% of the patients in the data set have more than 33 appointments during the calibration period (Fig. 3).

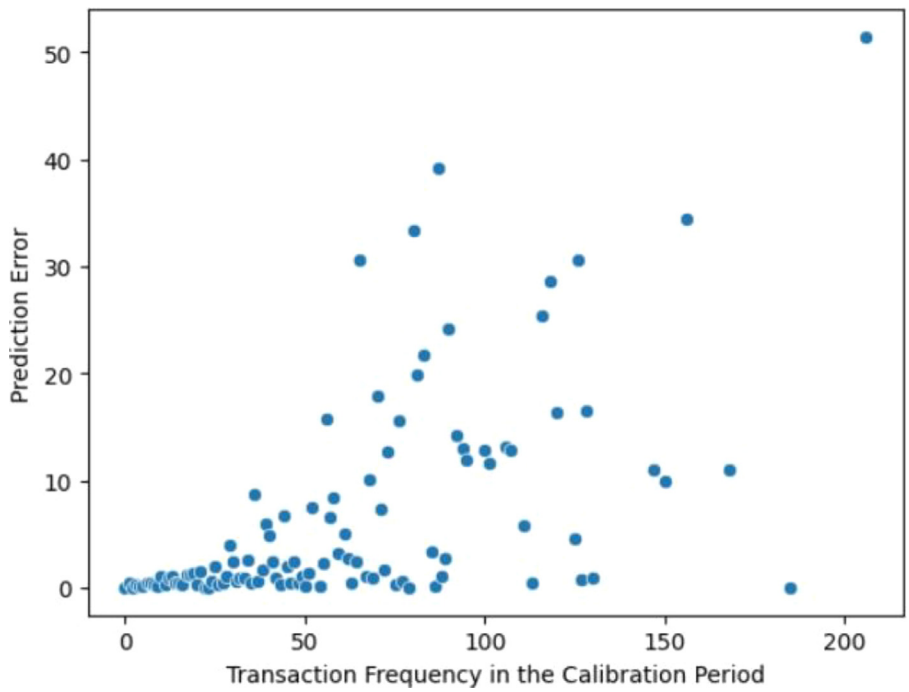

Fig. 2. Prediction error vs appointments in the calibration period.

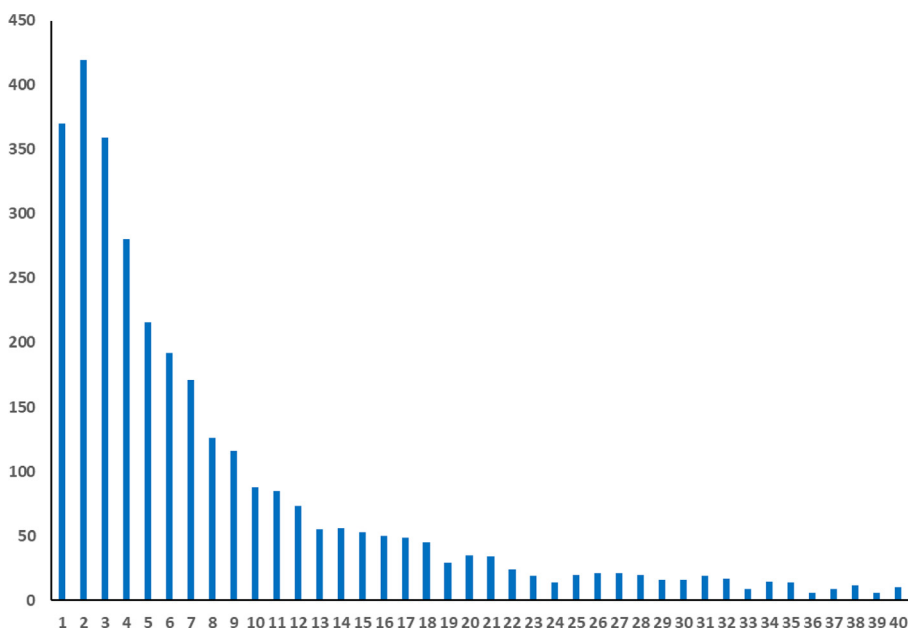

Fig. 3. Distribution of the frequency up to 50 in the calibration period (N=3383).

4.4. Estimating the probability of being alive

In the context of CLV investigation, the probability of being alive is the likelihood that a customer is still active at a specific time. For this study, being alive means that a patient remains engaged and will complete appointments within the holdout period. It helps us to understand whether patients are likely to continue with a clinic based on their past behavior. A higher probability means that the patient is more likely to remain. On the other hand, a lower likelihood suggests a higher chance that the patient will leave the clinic for good.

The BG/NBD model is used to estimate the probability of being alive [16,45]. Fig. 4 shows the result for a specific patient who had an initial appointment on 7 December 2021, followed by additional appointments later that month and the subsequent months (represented by red dotted vertical lines). The blue line represents the calculated probability that this patient is alive.

Following the patient's last appointment in January 2022, there has been a noticeable decline in the belief that the patient will continue to engage with the clinic. This decline has been rapid, considering the patient's past behavior. Six months after the last appointment, the probability that this patient is still alive has decreased from 0.9 to 0.2.

The rate of decline of the probability varies depending on the frequency of appointments. This is further depicted in Fig. 5. For this research, the estimated probabilities are stored at the patient level for further evaluation and comparison with those derived using the ML approach below.

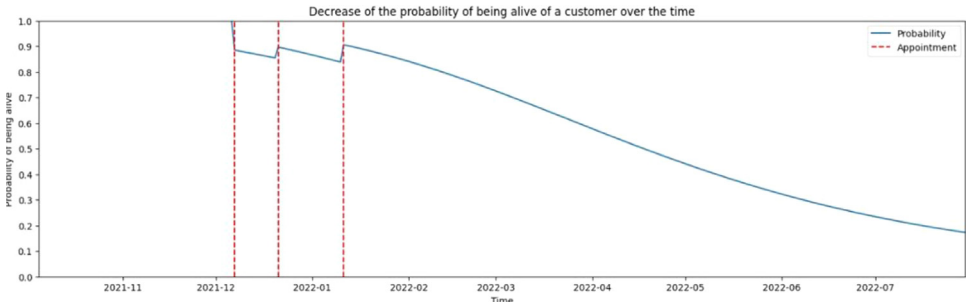

Fig. 4. Probability of being alive Over time - Patient 1.

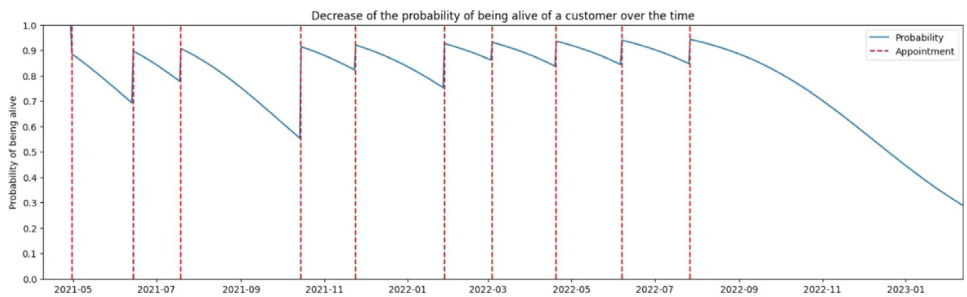

Fig. 5. Probability of being alive Over time - Patient 2.

4.5. Prediction of patient lifetime value

In this section, we present the results of the predicted CLV using the BG/NBD model and the Gamma–Gamma–Gamma submodel [45]. An additional variable, the monetary value, was included in the calculations. This monetary value can represent profit, revenue, or any value that is calculated consistently for each patient. The predicted CLV would then reflect the same value for the patients over their "lifetime" with the business.

According to [45], the monetary value, if computed from one's own data set, should be "the mean of a given customer's value, not the sum". Therefore, the monetary value used in the modeling process should be the average revenue (or profit) per transaction calculated at the customer level. In our case, since the data set used for this research does not contain billing or revenue data for each appointment, the monetary value was calculated by multiplying the average number of hours billed by each appointment by $ 200 per hour, a reasonable amount for billing. It is important to note that the average does not include the assessment appointments. The Gamma–Gamma–Gamma submodel assumes that "there is no relationship between the monetary value and the purchase frequency". [45]. Therefore, one would need to check whether this is a correlation between the two variables in practice (using a Pearson correlation coefficient, for example). Again, based on our understanding of the clinic's business model, we believe that the average number of hours billed to a patient for each appointment has very little to do with the number of patient appointments over time. Therefore, the assumption should be valid. This is confirmed by the correlation between the average number of hours and the number of appointments being −0.2.

Using the Lifetimes package, the predicted value of each patient was calculated for the holdout period. As expected, the predicted value distribution is skewed to the right, with many patients having zero values. To better illustrate this situation, its empirical inverse distribution is constructed and presented in Fig. 6. According to this figure, 50% of the patients have a value of $363 or less, while 90% of the patients have a value of $1,904 or less.

Note that the predicted value presented here is not CLV per se. It is calculated and considered here as a demonstration of concepts. If a discount rate is specified, we can use the Python package to calculate the CLV.

To demonstrate, three patients were randomly selected and the appointment histories and their CLV are shown in Fig. 7.

Patient 3 had the highest lifetime value, indicating high appointment frequency, long tenure, and low recency. Patient 2610 had moderate frequency, high recency and similar tenure, but lower consistency of appointments in recent years, resulting in a relatively low lifetime value. Patient 5218 had low appointment frequency, tenure, and recency. However, the patient has a higher lifetime value due to a higher probability of being alive than Patient 2610. Consistent with our intuition, the key drivers for high lifetime value are low recency, high frequency, and long tenure.

5. Predictive modeling via machine learning algorithms

This section describes a viable approach using ML algorithms to determine whether a patient is alive. The results of this development are compared with those obtained using probability models. Given that the ML algorithms allow for the inclusion of exogenous variables such as a patient's demographics into the modeling process, conceptually, these algorithms will have the potential to give better results than those derived using the probability models.

As mentioned above, a patient is considered "alive" if they have a completed appointment in the holdout period; otherwise, the patient would be classified as "Not Alive". Furthermore, around 80% of the records in the research data set were randomly selected to form the training data set for the ML modeling exercise. The remaining 20% of the data set was used as the testing data set. Some patients were excluded in the data set for the ML modeling exercise if their age or gender information was missing or the appointment date was beyond 30 August 2023. This ensures that the data used in the development process has no missing features. To ensure a valid comparison with the probability model, appointments that were not completed were also excluded. Only 2505 patients and 38,667 records met these criteria. Consequently, 585 of the 2004 patients in the training dataset and 141 of the 501 patients in the testing data set are labeled "Alive".

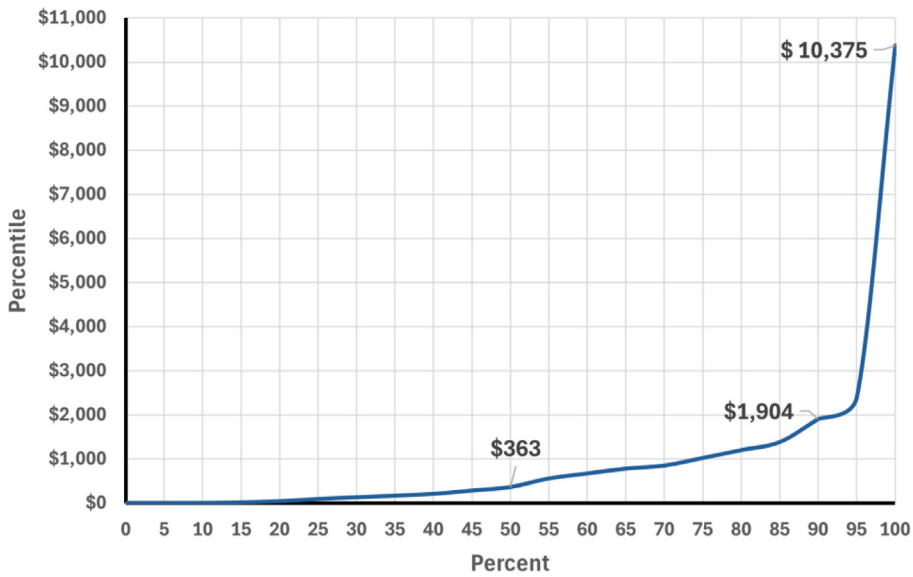

Fig. 6. The Empirical Inverse Distribution Function of Predicted Value of Patients in the Next Twelve Months.

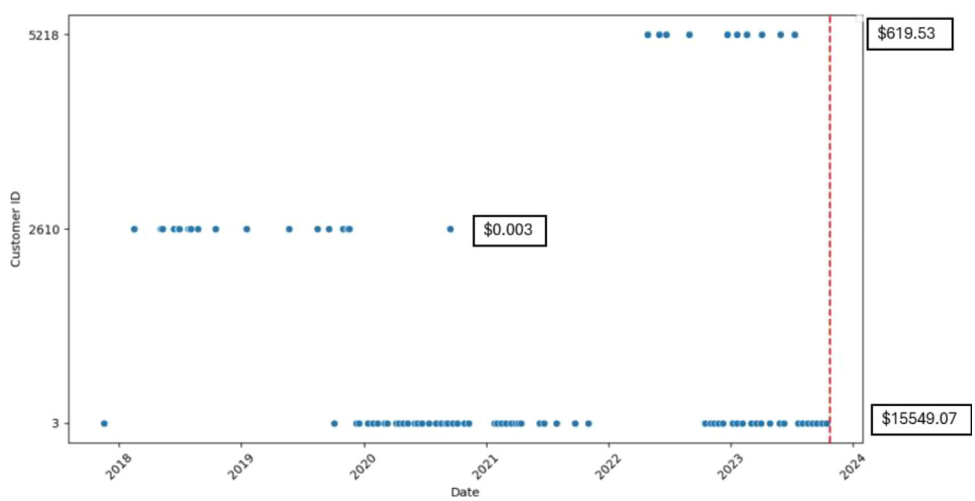

Fig. 7. Appointment History and CLV for Three Randomly Selected Patients.

5.1. Feature selection and engineering

As presented above, the research data set contains two exogenous variables: gender and age of a patient. In addition, frequency, recency, and tenure are also included as features in the modeling process.

The gender variable was recorded as "M" or "F". It was transformed into a binary (0,1) variable as input to the models. On the other hand, the age variable is used as is with no standardization or normalization applied.

The distribution of the gender variable including in the ML modeling process is presented in Table 5.1. Its slight imbalance will not affect the accuracy of the ML models to be developed below. Table 5.1 provide a statistical summary of the Age variable,

Gender	Count	Percent
F	1,208	48.2%
M	1,297	51.8%
Total	2,505	

5.2. Predictive modeling

In the following, we introduce the various aspects of the predicting modeling exercise and the key results.

5.2.1. Machine learning algorithms

As indicated in the literature review, the application of ML algorithms to the problem at hand is relatively new. As illustrations, we will apply various ML algorithms with different structures and approaches to gauge their effectiveness. Sina Mirabdolbaghi and Amiri [50] is a good reference for the recent research in this area. Other research using ML algorithms for this problem includes [11,42,51], and [52].

Statistic	Age
Mean	32.4
St. Dev.	14.1
Minimum	9
First Quartile	21
Median	30
Third Quartile	42
Maximum	82

Logistic regression (LR) is applied to the classification problem as the only statistical algorithm used in the research. This technique will act as the baseline, even though its effectiveness is not expected to exceed that of the other ML algorithms.

In this research, two gradient boosting algorithms, XGBoost (XGB) and LightGBM, are employed. XGB is well-documented to be adequate for classification problems. It can also be used to predict the CLV [11]. LightGBM [50,53] has emerged as a strong competitor of XGB for training time and accuracy. In this research, the Genetic Optimization (GO-LightGBM) [54] and Bayesian Optimization (BO-LightGBM) [55] algorithms are used to identify the optimal set of hyperparameters for the LightGBM model.

Two other ML algorithms, the support vector machine (SVM) and multilayer perceptron (MLP), are also included, as they are very different in approach and structure with the other techniques.

5.2.2. Evaluation of model performance

Various performance metrics can be used to measure the effectiveness of the ML algorithms deployed in this research. Most of these metrics are calculated using the confusion matrix. This matrix shows the number of predictions for each class that are correct and those that are incorrect. For a binary classification problem, there are four numbers in the matrix.

- True positives (TP) - The number of "positive" observations the model correctly predicted.
- False-positive (FP) - The number of "negative" observations the model incorrectly predicted as "positive".
- True negative (TN) - The number of "negative" observations the model correctly predicted.
- False-negative (FN) - The number of "positive" observations the model incorrectly predicted as "negative".

Using these four numbers, we define several performance metrics as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

Accuracy is used to gauge the overall accuracy of the classification. It can be presented as a percentage.

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

Precision is used to gauge how well the classification model correctly identifies the positive class. It can also be presented as a percentage.

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

Recall measures how well the classification model correctly identifies all the positive observations in the data set. It can be presented as a percentage. However, it does not provide information on false positives. One can evaluate precision and recall together through a precision–recall curve.

$$F1 = \frac{2 * Precision * Recall}{Precision + Recall} \tag{4}$$

Mathematically, the F1 score is the harmonic mean of precision and recall. It is easy to see that the value of F1 is between 0 and 1 and, therefore, can also be presented as a percentage.

We will focus on the Matthews Correlation Coefficient (MCC) [56]. It can be calculated from the confusion matrix as follows [57]:

$$MCC = \frac{TP * TN - FP * FN}{(TP + FP)(TP + FN)(TN + FP)(TN + FN)} \tag{5}$$

According to several papers written by Chicco and others [57–62], as a metric, MCC is a better measure for classification problems like ours. The concern that MCC may not perform as well for imbalanced data [63] is not warranted in our case.

Table 2
Table of optimized hyper-parameters.

Algorithm	Hyper-parameters
LR	regularizationstrength=0.1, penalty="L2", solver="saga"
GO-LightGBM	max depth=4, Cosample by tree=1, Learning Rate=0.03, # estimators=200, subsample= 0.5, random state=4
BO-LightGBM	max depth=2, Cosample by tree=0.5, Learning Rate=0.07, # estimators=200, random state=4, subsample= 0.5
XGB	maximum depth=2, learning rate = 0.2, # estimators = 100, random state=4
SVM	C=100, epsilon=0.1
MLP	activation="logistic", alpha=0.001, hidden layer sizes=(50, 50), max iter=500, random state=4

Table 3
Modeling results.

Actual/Predicted	LR	MLP	LightGBM	XGB	SVM	Voting
TP	342	325	340	333	335	338
TN	64	84	78	81	63	96
FP	77	57	63	60	78	45
FN	18	35	20	27	25	22
MCC	49.1%	52.7%	56.4%	54.6%	44.6%	65.7%
Accuracy	81.0%	81.6%	83.4%	82.6%	79.4%	86.6%
Precision	81.6%	85.1%	84.4%	84.7%	81.1%	88.3%
Recall	95.0%	90.3%	94.4%	92.5%	93.1%	93.9%
F1	87.8%	87.6%	89.1%	88.4%	86.7%	91.0%

Other measures generated and used in the evaluation include the Area under the ROC Curve (AUC). See, for example, [64] for details on calculating this metric. This is a popular classification performance metric, but its use could be misleading [65]. For a reason given by [57], measures such as Cohen's Kappa and Brier's score will be omitted here.

5.3. Modeling results

The results of the modeling exercise are documented below. Note that the same feature set has been used for every algorithm. Except for GO-LightGBM and BO-LightGBM, the ML models are created through a hyperparameter tuning process using grid search and five-fold cross-validation. The GO-LightGBM and BO-LightGBM algorithms use their respective optimization processes for fine-tuning the hyperparameters. Unless otherwise stated, the models were developed through a Python library for the underlying algorithm. Table 2 summarizes the optimized hyperparameters used to create the models.

Hard majority voting is an ensemble technique that can be used to improve performance by combining several classifiers [66–69] and is readily available in Python [70]. Two options are available in a voting algorithm. If the soft-voting option is used, the predicted class gives the maximum of the predicted probabilities of the classifiers involved. The hard vote option is much simpler because the predicted class label is based on the majority rule of the individual classifiers. For this exercise, the hard-voting algorithm is implemented by combining the GO-LightGBM, XGBoost, and SVR classifiers to create a new one. As we can see in the following, this voting classifier performs better than the three it combines.

A confusion matrix is generated for each model and is presented in Table 3. The performance metrics described in Section 5.2.2 were also calculated and summarized in the table.

We will omit results from the BO-LightGBM model here as they are very close to those from the GO-LightGBM algorithm. The GO-LightGBM algorithm will be denoted LightGBM going forward. As expected, LR produces inferior results across most performance metrics and should not be a choice in the future. The same can be said about MLP, although with more elaborate structures, deep neural networks could still be a modeling option for future research.

Table 4
Comparing ML models with the probability model.

Metric	LightGBM	XGB	SVM	Voting	PM
MCC	56.4%	54.6%	44.6%	65.7%	70.1%
Accuracy	83.4%	82.6%	79.4%	86.6%	87.4%
Precision	84.4%	84.7%	81.1%	88.3%	93.0%
Recall	94.4%	92.5%	93.1%	93.9%	89.2%
F1	89.1%	88.4%	86.7%	91.0%	91.1%

Unsurprisingly, LightGBM and XGB are the two viable options. In this case, the size of the dataset and, therefore, the processing speed are not concerns. The choice will therefore come down to the accuracy achieved by the classifiers. The LightGBM is the narrow winner in this regard. Furthermore, it is encouraging that the voting algorithm produces the best ML model for all performance measures except recall.

The MCC for the LightGBM and XBG models is about 55%, which is considered good. The voting model improves this metric further to 65.7%. The next section compares these results with those from the probability model to gauge their efficacy.

It is worth noting that other ensemble methods, such as the weighted voting algorithm, were also considered. A weighted model developed for this research used the accuracy of the three basic ML models as the weights. The results of this model are the same as those of the majority-voting model, and therefore, they are omitted here.

6. Comparing the two approaches

Results from the ML approach will now be compared with those of the probability model.

The basic ideas in this comparison are as follows.

- If there are (completed) appointments between 4 August 2022, and 31 August 31 2023, a patient is labeled “alive”.
- Use the appointment data up to 3 August 2022 to develop the probability models as above
- Use the training data set described in Section 5 for developing the ML models
- Likewise, use the testing data set for evaluating the results

The first set of comparisons is to see which approach could produce models that are more effective in classifying the patients as “alive” correctly on 3 August 2023. Given that the probability model produces a probability that a patient is “alive”, we need to use a classification threshold (T), that is, if the probability of being alive is higher than the threshold, the patient is classified as “alive”. For this evaluation, the threshold is set to 0.5. Table 4 is created from the resulting confusion matrix after the models are applied to the testing data:

The probability model (PM) performs better than the best ML model in all but one performance metric (recall). This result is remarkable because the ML models were built using additional information (the two demographic variables).

The probability model presented in Table 4 used a threshold of 0.5, that is, if the probability of being “alive” is greater than 0.5, the patient is classified as “alive”. One would question whether a change of the threshold would have an impact on the performance of the probability model. Table 5 gives the answers to this question with different threshold values used (in brackets):

It seems that the “optimal” threshold was 0.7. It produced an accuracy rate of 88.4%, an F1 of 92.1%, and a slightly lower level of precision.

The total number of patient appointments in the twelve months ending 4 August 2023 is calculated to compare the accuracy of predicting the value of patients using the probability model and the ML models. This value is used as a proxy for the value of patients for training the ML models. The test data set is then used to perform the comparisons.

Table 5
Performance of the probability model using different threshold values.

Metric	LightGBM	Voting	PM (0.5)	PM (0.6)	PM (0.7)	PM (0.75)
MCC	56.4%	65.7%	70.1%	69.1%	70.8%	69.5%
Accuracy	83.4%	86.6%	87.4%	87.4%	88.4%	88.0%
Precision	84.4%	88.3%	93.0%	91.6%	90.8%	89.7%
Recall	94.4%	93.9%	89.2%	90.8%	93.3%	94.2%
F1	89.1%	91.0%	91.1%	91.2%	92.1%	91.9%

Table 6
Prediction accuracy.

Model	MAE	RMSE
LightGBM	4.21	10.52
XGB	4.57	12.63
SVM	4.30	10.45
Voting	4.32	11.13
PM	2.40	5.67

The features used in this modeling exercise are the same as those used to develop the above classifiers. Note that there are many zeros in the training and testing data set. We believe that a more appropriate approach is treating the data as zero-inflated or one from a hurdle model [71–73]. The comparison process is therefore summarized as follows.

1. Use SVR, XGBoost, and LightGBM to classify the training data set into two groups: alive or not alive and create three sets of results.
2. For the results from SVR, XGBoost, and LightGBM, use the actual “alive” group in the training dataset to train a machine learning algorithm to predict the number of appointments.
3. Use the ML classifiers and the voting classifiers to classify the testing data set.
4. Apply the ML and the Voting estimator to predict the number of appointments in the testing data set. Note that the voting estimator is 0 if the majority vote is “not alive”. If the majority vote is “alive”, it is the average of the predictions from those classifiers that voted “alive”.
5. Compare the results with the probability model developed in Section 4.

Using the above approach, each classifier’s predicted number of appointments in the testing data set will have TP number of zeros. The results of these estimators are compared with those from the probability model using the Root Mean Square Error (RMSE) and the Mean Absolute Error (MAE). A formula for these two performance metrics can be found in many statistics texts. See, for Example, [74]. Note that another popular metric, Mean Absolute Percent Error (MAPE), cannot be used here as the data set contains a significant number of 0’s.

Table 6 compares the prediction of the value of patients in the holdout period using the probability model and the various ML models:

Again, the probability model performs favorably compared to the ML models. Furthermore, the voting algorithm does not improve on the underlying ML algorithms as it is an average.

7. Discussions

Although the data set used for this research is not small, the deployed ML algorithms could benefit from a larger data set. Future applications should focus on acquiring a patient data set with more patients, more transactions, more demographic features, and, ideally, a complete history of the past. Temporal patterns and behavioral indicators such as interappointment times, clinical variables such as broad categorical of clinical issues, and socioeconomic data such as whether the visits are paid for by a third party (insurance or government) could

be used as independent variables in the modeling process and could significantly enhance the accuracy of models.

The probability model implemented in this research compares favorably with the ML models described above. This is probably due to the relatively strong distribution assumptions (Section 2 in [16]) required for the models. An essential step before using this probability model is to verify that the assumptions made in the probability model are valid.

More detailed analysis on the regularity of appointments would also be helpful [27,28,75,76]. If a measure of regularity is assured (this is possible for a health service portfolio such as the one analyzed), we could use other probability-based algorithms, for example, BTYDplus [29], to see whether the result could be further improved.

The process used to estimate the number of patient appointments using ML algorithms is heuristic in nature. More research is required to establish a framework for using ML algorithms for estimation with zero-inflated data beyond the work completed in [72] and in [73].

8. Conclusions

In this research, we apply the probability model developed under the RFM paradigm to classify a patient portfolio from a health service business and predict the lifetime value of a patient. This model is compared with several machine learning models in terms of classification and prediction accuracy. The probability model performs well in these comparisons. This is likely due to the assumption required in the probability calculations. Therefore, an essential step in applying these software approaches is to verify the validity of the key assumption of regularity. The results of the research can be solidified with a larger dataset with more demographic features. Furthermore, there is still much room to improve the application of ML models to calculate customer lifetime values. Developing specific ML models for dealing with zero-inflated data will be most helpful in this regard.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Albert Wong reports financial support was provided by Natural Sciences and Engineering Research Council of Canada. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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